Northern Ontario Proposed Graphite Mine Impact Statement

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1. Introduction

In 2021, the Canadian federal government identified 34 essential critical minerals for national security and improvement of Canada's position in global supply chains [1]. Graphite is an especially important mineral as it is the main material in lithium-ion batteries, which are critical for global electrification efforts [2]. To improve Canada's political and economic position, it is important to develop the country's own graphite mines. Mining projects fall under the Physical Activities Regulation and thus require a federal impact assessment under the Impact Assessment Act 2019 [3]. This report will outline the proposed graphite mine's key valued components, including their importance and magnitude, baseline assessments, predicted impacts, and mitigation and monitoring strategies.

1.1 Location – Our project proposes the development of an open-pit graphite mine in Northern Ontario, based on the Albany Graphite Deposit near Hearst (see Appendix A). The site lies within the Hudson Bay–James Bay Lowlands, about 30 km north of the Trans-Canada Highway and 35 km northwest of the Constance Lake First Nation [4]. It is close to the Nagagami and Pitopiko Rivers and located within the Pagwachuan range of woodland caribou habitat. This location was selected for its high-purity graphite suitable for graphene production and its proximity to existing infrastructure such as Highway 11 and power lines. [5]

1.2 Stakeholders – This project will affect the nearby town of Hearst (60 km southeast of the project site) and the Constance Lake First Nation (35 km southeast of the project site) due to its proximity to these communities (see Appendix B). Hearst is a small Franco-Ontarian town with a population of about 5000 people. Their government is composed of a mayor and six elected councilors [6]. Its economy was established about 100 years ago with the arrival of the Canadian National Railway and has historically been driven by forestry. The town may be significantly impacted by the project, alongside resident wellbeing, health, and recreational activities [7]. The Constance Lake First Nation is a community of approximately 1600 members of Cree, Oji-Cree, and Ojibway descent. About 820 of them live on the reserves, Constance Lake 92, and English River 66, which cover 7686 acres of land near the project site and are part of Treaty 9. Their government now consists of an elected chief and six elected councilors [8]. They engage in fishing, hunting, and trapping in the land and waters near the project [9]. Overall, the project's impacts on the physical environment, economic landscape, and social dynamics will concern not

only the activities, health, and lived experiences of these local communities and Indigenous groups, but also that of the greater public through wide-spread environmental effects, such as GHG emissions. There are also several regulatory bodies which may have interest in this project (see Appendix C).

1.3 Boundaries – Jurisdictional boundaries dictate permits and approvals at various levels. Federal approvals aim to identify and evaluate the significance of possible environmental effects [10]. Permits include Authorization for Works Affecting Fish Habitat under the Fisheries Act [11] and License for a Factory and Magazine for Explosives under the Explosives Act [12]. Provincial approvals are set by various ministries, such as the Ontario Ministry of Natural Resources. Permits/approvals include but are not limited to Approval for Air Emissions under the Environmental Protection Act [13], and the Mine Closure Plan under the Mining Act [14].

Spatial boundaries – The following boundaries define ranges for impact assessment [15], [16]

- **The project area** includes the footprint and immediate vicinity of the project area. This includes the mining site/facility and roads. Activities like construction are directly performed in this area.
- **The local assessment area** is the area to the extent of environmental effects on a given valued component and varies based on the VC under consideration.
- **The regional assessment area** is the area where cumulative effects and impacts of potential accidents are studied.

Temporal Boundaries — Impacts from all phases of the project from construction to reclamation are considered. Construction is expected to last under 2 years, and the lifetime of the mine is expected to be 22 years [17]. Progressive rehabilitation is expected to start 10 years into the project, closure restoration to happen for 1-2 years after mining completion, and post closure monitoring for 6-10 years.[15], [16]

1.4 Alternatives – The International Energy Agency identifies lithium, nickel, cobalt, and manganese as key materials for batteries [18]. As seen in Appendix D Figure D2, there are no known nickel deposits at the site, and manganese is uncommon in Northern Ontario [19]. However, lithium and cobalt deposits are nearby, thus these alternatives will be assessed (Appendix D Figure D1, Figure D3). Lithium mining is water-intensive, can damage surrounding

land, and often leads to toxic waterways [20]. Cobalt, mainly mined in the Republic of Congo, is linked with water and fish contamination, toxic air quality, and human health risks like cobalt toxicity [21]. In comparison, graphite is less resource-intensive, and with proper tailing management, water contamination can be contained. Graphite is also non-toxic, and will not harm waterways, fish, or air in limited quantities [22]. Overall, graphite mining is the most economic and safest option. Different designs were evaluated. While open pit mining has a greater ecological footprint, it is more cost effective, safe, and scalable than underground mines [23]. Alternate locations were also considered. As seen in Appendix D Figure D4, the chosen site is the northernmost confirmed graphite deposit, providing the most development benefits for Northern Ontario. Nearby graphite occurrences would require exploration and drilling, inducing additional environmental impacts and considerations.

2. Valued Components

The valued components of the proposed graphite mine are employment, wildlife, traditional culture and land use, soil quality, water quality, navigation, vegetation, air quality, noise and vibration, greenhouse gas (GHG) emissions, essential services, transportation, public utilities, and residential communities and housing. To evaluate the relative importance and magnitude for environmental, social, economic, and health valued components, a Leopold matrix was used. Each valued component is given a magnitude and importance value from –10 to 10 for each impact activity (see scale used in Appendix E, Table E5). The magnitude considers the activity's impact on stakeholders, including the duration, frequency, likelihood, and spatial extent of the impact. The importance assesses the duration of benefit for positive impacts, and the extent of mitigation possible for negative impacts. Impact matrix scores for each valued component are shown in Appendix E, with the valued component with the highest overall score highlighted in red. The valued components with the greatest score in their section and thus chosen for further analysis are air quality, water quality, caribou, and transportation.

3. Air quality

3.1 Magnitude and importance – Air quality is a key health consideration in the proposed graphite mine due to its impact on human health. The mine's entire lifetime from excavation to reclamation has directly negative effects on air quality. Blasting releases particulate matter and pollutants such as CO, NO_x, and SO_2 due to the use of explosives [24]. Mine waste disposed in

tailing dams, mineral processing and refining, and mineral transportation with trucks can all generate dust as PM_{10} and $PM_{2.5}$ [25]. At high concentrations, dust and pollutants can be harmful to workers and nearby communities. While particulate and gas release can be mitigated to a certain extent, it requires strict monitoring programs to ensure safe working and living conditions. Air quality is impacted across a mine's lifetime, with potentially dangerous and harmful effects, thus the impact magnitude is moderate. The proposed mine is situated near several Indigenous communities and will house a mining town with workers and families [4]. Additionally, the land is a known woodland caribou habitat and nearby the proposed Magino mine [4]. Thus, impacts on air quality will impact community members and wildlife, and can have cumulative effects with future projects, making the importance of air quality impacts high.

3.2 Baseline assessments & impact prediction – To determine baseline conditions on the proposed mining site, a combination of onsite air quality measurements and regional data collection will be used. The National Air Pollution Surveillance (NAPS) program has various air pollutant data collection sites across Ontario [26]. Data on SO_2, CO, NO, NO_2, NO_x, and $PM_{2.5}$ concentrations will be taken from Thunder Bay, and data on PM_{10} will be taken from Sudbury [27]. On site, low volume sampling will be used to find existing $PM_{2.5}$ and PM_{10} concentrations [28]. These on-site observations will be used to confirm the regional data representation of Northern Ontario conditions, as many surveillance stations are situated far from the site.

To predict the impact of air quality, portable wind tunnels will be employed on site prior to construction to perform dust emission rate measurements from different materials at varying wind and surface conditions [29]. These results will be used for air dispersion models, a proven technique for predicting air quality compliance in many jurisdictions, to predict contaminant transport and impact [29]. The software to be used is AERMOD and CALPUFF as both are industry standard tools [29]. There are also innovative techniques within particulate emission impact prediction that employ machine learning tools such as K-means clustering and random forest algorithms to make predictions [30]. Predictions from model-based techniques will be compared with input from experts on air quality impact in mining, traditional knowledge from nearby Indigenous communities, and community members residing near operational mines in Ontario. To estimate cumulative effects with the proposed Magino Mine, contaminant interactions will be simulated within air dispersion models to observe resulting effects.

3.3 Mitigation – To minimize the predicted impacts of air quality from mining activities, several mitigation strategies will be implemented. The first mitigation strategy will be to set blasting procedures to limit how long blasting material can sit out, and delegate blasting times to days with best air dispersion conditions to minimize unsafe particulate and pollutant concentrations [28]. Next, tailings and other material waste will be used for backfilling whenever possible to reduce time of exposure for dust generation [31]. Materials will be handled indoors as much as possible to reduce particulate emission, and all buildings will be equipped with ventilation systems with proper filtration to ensure safe working conditions [32]. Dust from transportation will be minimized using strategic mine layouts to minimize distance needed to be covered by haul trucks [28]. To control dust emissions around the mine, water sprays, sealants, and chemical suppressants will be used on roads and mineral stockpiles [32]. All workers will be trained to use dust combating techniques, and all equipment will be well-maintained [33]. A dust management plan will be developed following Ontario's "Management Approaches for Industrial Fugitive Dust Sources" before starting construction [34].

3.4 Monitoring – To monitor the impacts of mine emissions, continuous particulate matter monitors will be placed around the perimeter of the mine site to measure PM_{10} and $PM_{2.5}$ concentrations alongside CO, NO_x, and SO_2 monitors to measure ambient pollutant concentrations (see Appendix G for examples). To assess the effectiveness of proposed mitigation strategies, regional air quality data from the Sudbury and Thunder Bay NAPS program sites will be used as control sites to compare against on-site measurements. If concentrations of contaminants exceed air dispersion model predictions and cannot be explained by regional trends, mitigation programs will be reassessed and adjusted accordingly. Moreover, if contaminant concentrations surpass Ontario's ambient air quality concentration guidelines (see Appendix F for values), alarms will go off to notify workers of unsafe air quality conditions and to proceed with established emergency procedures. Analysis of dangerous concentrations will be conducted to inform changes to mitigation strategies.

4. Surface water quality

4.1 Magnitude and importance – Water quality is a major valued component from both a stakeholder perspective and regulatory perspective. It is an indicator of human and environmental health as well as a health indicator of aquatic and terrestrial resources and systems

[35]. It is closely linked to the traditional activities of the Constance Lake First Nation who use the nearby rivers for fishing. Water quality is also regulated under the Environmental Protection Act and under mining-specific regulations such as Metal and Diamond Mining Effluent Regulations (SOR/2002-222) [36]. The project is situated near the Nagagami and Pitopiko river that feed into James Bay, where the water quality may be impacted by new mining infrastructure, preparation of terrain, construction work, circulation of vehicles, runoff from the mine and other mining activities. The two main impacts of focus are the decrease in water quality due to the discharge of mine water and the release of ammonia due to the use of explosives. Studies have shown that graphite flakes are toxic towards daphnia (water fleas) [37], and traces of iron sulfide minerals, pyrite, and pyrrhotite, commonly present in graphite deposits, can cause acid-rock drainage if exposed to water in waste rock or tailings. Runoff, wastewater, and mine drainage, if not properly used, treated, and discharged, could impact water quality, in turn harming aquatic life, land animals, and people that use this water as a source of drinking water [38]. Ammonia, when carried into aquatic ecosystems through mine runoff, can harm fish by affecting their tissue development, damaging their eyes, gills, and skin, and causing hyperplasia. It can also increase the growth of toxic algal blooms, which reduces levels of dissolved oxygen for aquatic life [39].

4.2 Baseline assessments – To measure baseline surface water quality, four different types of parameters will be assessed. Physical parameters, including pH, temperature, turbidity, total suspended solids (TSS), specific conductivity, and hardness will be measured. Acidity (pH) and hardness can affect the toxicity of metals. Conductivity and turbidity can be an indication of runoff or high levels of TSS. A high turbidity or TSS could affect light penetration through the water, which could reduce plant growth and smother benthic habitats. Total and dissolved metals will help establish a baseline of metal concentrations prior to mine operations, which can increase in concentration if waste is not properly managed. Nutrients, such as nitrogen (ammonia, nitrate, nitrite) will help establish trophic status before any potential discharge from project-related activities. Finally, anion content will be measured, including sulphate, which is a key indicator of mine drainage and the impact of tailings [40]. There will be some seasonal variability to many of these water parameters. Sampling will be conducted over one or more years to capture high flow, low flow, rainy and winter conditions to effectively distinguish natural variations from mining impacts. Sampling locations serving as reference sites will be set in the upstream parts of the Pitopiko and Nagagami rivers, and Nagagami Lake. These samples

will be compared with downstream sites, including the direct discharge point of the mine, the receiving sites in the Nagagami and Pitopiko river, and the mouth of James Bay [41].

4.3 Mitigation – The best way to mitigate impacts on surface water quality is to first reduce the quantity of water that needs to be treated and discharged to the surrounding environment. Water from outside the site will be deviated to avoid contamination. Water within the mine will be reused as much as possible after being treated [15]. All processed water, mine drainage water, runoff from the mine, and sewage, will be collected in water basins to allow treatment in accordance with provincial and federal guidelines [42]. This system will address the neutralization of the acid mine water with the use of sodium bicarbonate, the removal of fine particles from runoff water with the use of a flocculant and filtration with Geotubes, and the removal of potentially leached metals using a coagulant and Geotube filtering. The flocculant and coagulant will settle with the particles at the bottom of the basin or remain with the filtered particles in the Geotubes [43]. Both methylisobutyl carbinol (MIBC) and potassium amyl xanthate (PAX), used in the desulphurization of tailings, will primarily be carried with the solids in water. The remains will decompose over time. MIBC can decompose by 94% within 20 days in the basins and PAX has a half-life of four days in water. The collection basins will be built to withstand harsh weather conditions [15]. To minimize the ammonia that could end up in mine water due to the use of explosives, we will use emulsion explosives, which dissolve in water much less easily. As specified above, the mine water will be directed to the collection basins and treated. The water treatment plant will not specifically treat ammonia. Effective management of explosives and natural degradation should be sufficient to minimize ammonia levels [15].

4.4 Monitoring and Follow-up – During the construction, extraction, and decommissioning of the project, monitoring will be done to ensure compliance with the Metal and Diamond Mining Effluent Regulations (SOR/2002-222). Deleterious substances, pH and flow rate of the rivers will be measured in the reference points and affected bodies, so that they don't surpass the maximum allowed concentrations (see Appendix H). Rainbow trout (96-hour test) and daphnia magna (48-hour test) will be measured to test if the mine effluent is acutely toxic. The discharge point and the receiving water body will be monitored for site-specific water quality (hardness, alkalinity, electrical conductivity, dissolved oxygen), and biological monitoring will be done on fish populations and benthic invertebrates. The frequencies and methods for these measurements

are outlined (see Appendix I). Annual reports will be submitted by March 31 of each year summarizing the previous year's monitoring results. If any of the monitoring requirements fail to be met or if maximum thresholds (see Appendix H) are exceeded, the inspector will be notified immediately, and a response plan will be implemented [44].

5. Caribou

5.1 Importance of Caribou – Woodland caribou is an important resource for its environmental and socioeconomical values. They are identified as a 'Threatened Species' under the Species at Risk Act [45], and the project is situated in Cervid ecological zone A of Ontario's Cervid ecological framework [46], which is known caribou habitat. Caribou informs environmental impacts as they are a key indicator species for boreal habitats due to their high sensitivity to changes in habitat [47], [48]. Many First Nation groups hunt caribou for subsistence. The intangible (cultural, spiritual) values are hard to evaluate, as substitutes cannot replace caribou meat and materials [47]; however, based on mass harvested, hunting has an estimated value of $1600/person/year [10]. For these ecological and socioeconomic reasons, caribou is a highly important valued component.

5.2 Impacts – The project is expected to have two primary potential impacts on caribou population health: habitat loss, and sensory disturbances. Habitat loss decreases the quantity or quality of resources available for caribou. Loss of caribou habitat is expected from clearing vegetation for the footprint of the project, including the mining site, and roads which may impede caribou movement. Sensory disturbances are unnatural noises, smells, lights, or vibrations which can impact habitat access, calving rates, and hunting success [10], [16]. These disturbances come from various project activities, like construction, air traffic, road traffic, and blasting.

Impact Significance and Prediction — The project is expected to have a low impact on habitat loss, as it is limited to the project's physical footprint. The project area is highly predictable and will not increase throughout the lifetime of the project. The significance of sensory disturbances is expected to be moderate, as the disturbed area is larger than the physical footprint, and impacts on animal behaviour are harder to predict. A similar noise modelling study performed for heavy equipment using ISO 9613 suggested disturbances will reach up to 3km around the project zone, outside of which noise levels will be lower than 30dBA [10].

Baseline assessments will be performed to estimate caribou population demographics, study caribou movement, and evaluate habitat health, as informed by frameworks found in the Ontario Caribou Conservation Plan [49]. Aerial surveys will be performed to estimate caribou population size, density, and demographics. In particular, the recruitment rate is estimated by the number of calves spotted per 100 adult females. Radiotelemetry collar studies will inform survival rates of adult female caribou. Recruitment rates and female survival rates are used to model the intrinsic rate of growth, which predicts population trends and informs current risk.

5.3 Mitigation – Habitat loss can be minimized by careful site planning and restored through mine reclamation after decommissioning. A compact site design will minimize total footprint area. A minimum of 200m buffer zones will be placed around forest creeks which caribou and other wildlife are likely to use [10]. Existing roads will be used where possible, and special care will be taken when planning new roads to avoid severing paths between specialized habitats [10], [50]. Regularly spaced crossing points will be made to allow caribou to cross these roads [16]. Upon decommissioning the mine, the project land will be rehabilitated to varying degrees based on its project use. Some features like roads will be restored very closely to their natural state by revegetation through native seed mixes [10], [51]. Stockpiles will be revegetated with Jack pine trees which provide tree cover and support lichen growth. Lichen is a critical food source for caribou that usually grows in bogs and fens; however, stockpiles do not have the water drainage properties apt for a bog nor a fen [10]. Sensory disturbances can be minimized through suppressive equipment and regulatory policy. Mufflers and dust suppression devices on machinery will reduce noise and limit the spread of unnatural smells. Electricity from the grid will be used wherever possible to avoid loud generator sounds and smells. Wherever the use of generators and compressors is necessary, they will be contained inside noise-reducing structures or containers [10]. Policy implementations will reduce disturbances from road and air traffic. Aircraft such as helicopters will have a minimum 500 m altitude limit to minimize noise [16]. Speed limits and scheduled convoys will be implemented, which would reduce the frequency of road traffic disturbances.

5.4 Monitoring – Behavioural changes due to habitat loss can be monitored through continued radiotelemetry collar and aerial survey studies, such as detecting caribou migration failures. Critical choke points identified in these studies can be relieved by implementing crossing points

at these key areas. Habitat restoration success would be monitored following the completion of revegetation. Vegetation survival, growth, and coverage will be studied for ten years following the restoration [15], [51], with further studies on soil conditions if necessary to determine the cause of vegetative failures [51]. Camera traps can be placed on restored sites to verify that caribou reinhabit the habitat, and adaptive solutions can be implemented accordingly [52]. Noise monitoring will be performed using sound level meters to measure the noise disturbances near the project area. This will be performed near the start of each phase of the project and when any significant changes to noise emissions are expected. If the noise levels are above what was expected in the noise models, adaptive solutions to further reduce the noise will be implemented. Continued aerial surveying can measure the impact on recruitment rates [10], [52], [53]. Significant changes in recruitment rates may indicate high sensory disturbances, and may warrant changes to policies, like increasing the minimum flight altitude, or modifying convoy timings, paths and frequencies to avoid migratory seasons or frequently used caribou paths.

6. Transportation

6.1 Magnitude and importance – Highway 11 is not only essential to the movement of goods and services across Canada, but it is also the backbone of Western Canada's gateway to Ontario and the East. It serves as a vital lifeline for Indigenous and rural communities, resource development, national defense logistics, and emergency response [54]; however, the highway network in Northern Ontario mainly consists of two-lane roads that face frequent winter disruptions, poor visibility, and high collision risks which increase the likelihood of accidents [55], [56]. Developing a large-scale industrial project along this route could exacerbate existing vulnerabilities. The additional flow of ore and supply trucks would accelerate pavement wear, increase vibration and noise, and heighten collision risks, especially during winter conditions [57]. These pressures on Highway 11 are expected to manifest through heavier traffic loads that reduce road capacity, greater safety concerns and slower emergency response times, and more frequent disruptions to everyday community travel. Given the region's dependence on this single corridor for essential services and connectivity, the magnitude of transportation impacts is considered negative, and their overall importance is high.

6.2 Baseline assessments – To establish baseline transportation conditions for the proposed graphite mine, several data collection and analytical methods will be used. Traffic volume and

road capacity will first be assessed using traffic volume data from the Ontario Ministry of Transportation (MTO). This will help identify existing flow patterns and potential congestion points along Highway 11 and nearby access roads. Capacity estimation will incorporate Passenger Car Equivalent (PCE) factors to reflect the influence of heavy-truck acceleration and braking on traffic flow, ensuring that results better represent mixed-traffic conditions typical of mining operations [58]. Road safety and emergency response will also be evaluated by reviewing collision frequency, severity, and seasonal trends using data from the Ontario Road Safety Annual Report. Emergency response times will be analyzed through provincial and local EMS data to establish baseline travel intervals for emergency services along Highway 11. Comparative insights from U.S. studies on rural EMS delays will be referenced to contextualize accessibility challenges common to remote regions of Northern Ontario [59]. Finally, community access and mobility will be analyzed using GIS-based network modeling to simulate changes in travel time from residential areas to key destinations such as schools, hospitals, and grocery stores once mining transport begins. Household travel surveys and community interviews will complement the model by capturing local perceptions of accessibility and safety.

6.3 Mitigation – Capacity reliability analysis will be used to identify vulnerable highway segments and prioritize pavement reinforcement or rerouting strategies before mine operations begin. This analysis helps identify critical and vulnerable segments, allowing planners to prioritize pavement reinforcement, targeted widening, or alternative routing to maintain a safe and efficient level of service [60]. Preventive upgrades and maintenance will ensure that major transport corridors retain sufficient capacity to handle both mine-related and regional traffic. To address road safety and emergency response delays, key sections of Highway 11 will be upgraded to a 2+1 lane configuration to improve passing safety and reduce collision risks [56]. Seasonal snow and ice removal programs will maintain visibility and traction during winter, while driver training and wildlife detection systems will further reduce accident potential. The project will also contribute to regional road safety and emergency preparedness funds to strengthen community response capacity. Finally, to prevent disruptions to local mobility, heavy-truck haulage will be scheduled outside of peak commuting and school transport hours, minimizing conflicts with community traffic and ensuring continuous, safe access for nearby towns and Indigenous communities.

6.4 Monitoring – To monitor the effectiveness of mitigation strategies, both indirect and direct methods will be applied. For road capacity, Ontario MTO and municipal datasets will be analyzed to track long-term trends in traffic volume, congestion, and maintenance frequency. Travel time reliability indicators combined with Floating Car Data (FCD) will provide continuous, cost-effective monitoring of traffic flow across the mine's transport corridor. By analyzing vehicle speed and travel time patterns, FCD allows early detection of congestion or capacity decline and helps evaluate whether highway upgrades effectively sustain the expected level of service. [61] For road safety and emergency response, annual reviews of accidents and EMS response records will be compared with baseline data to confirm that risks remain under control. GPS tracking and on-site speed audits will verify compliance with safety protocols, with additional training or infrastructure improvements implemented if collision rates increase. For community access and mobility, regular meetings with nearby towns and First Nations, along with annual accessibility surveys, will collect feedback on travel times and safety perceptions. These findings will be validated using quantitative reliability metrics to ensure local connectivity remains unaffected.

7. Conclusion

This report evaluated the potential impacts of the proposed Northern Ontario graphite mine across four key valued components: air quality, water quality, wildlife, and transportation. It outlined a comprehensive approach for managing impacts that includes baseline assessments, impact prediction, mitigation, and monitoring. Moving forward, the project will implement the described mitigation and monitoring strategies to ensure issues are identified and addressed as soon as possible. This adaptive management approach ensures that environmental, social, and health impacts are minimized. The project's combination of proactivity, adaptability, and preparedness makes it well positioned to protect valued components while providing significant benefits to Ontario, and thus feasible and responsible to advance.

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Copper

Platinum group elements

(Ni-Cu-PGE). Available: https://www.ontario.ca/files/2024-10/mines-nickel-copper-platinum-group-elements-en-2024-10-11.pdf.

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Appendix A. Map of Albany Graphite Deposit

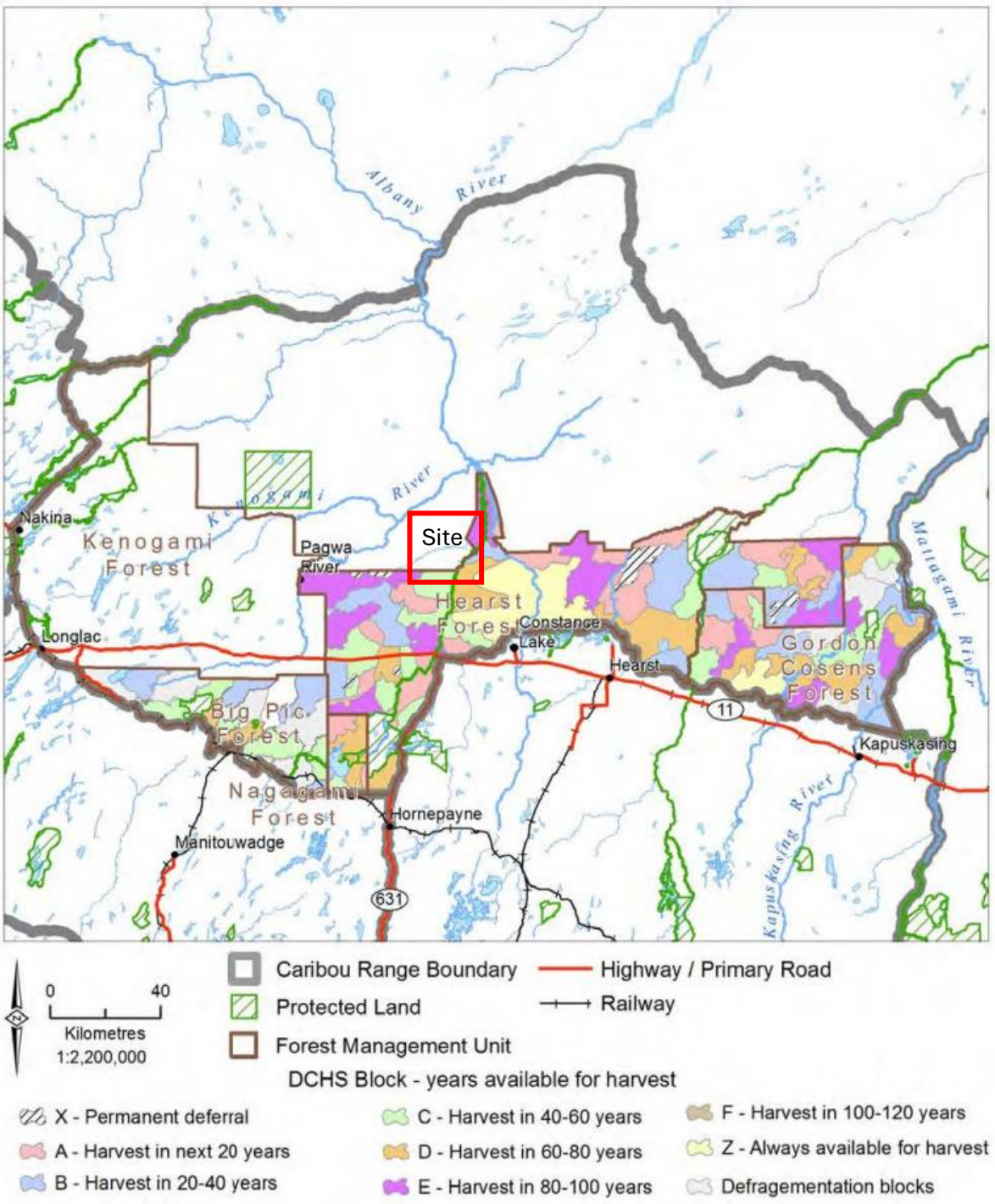


Figure A1. Map of proposed mine location showing existing caribou habitats, highways, protected land, railways, and forest management units

Appendix B. Location of Mine in Relation to Stakeholders

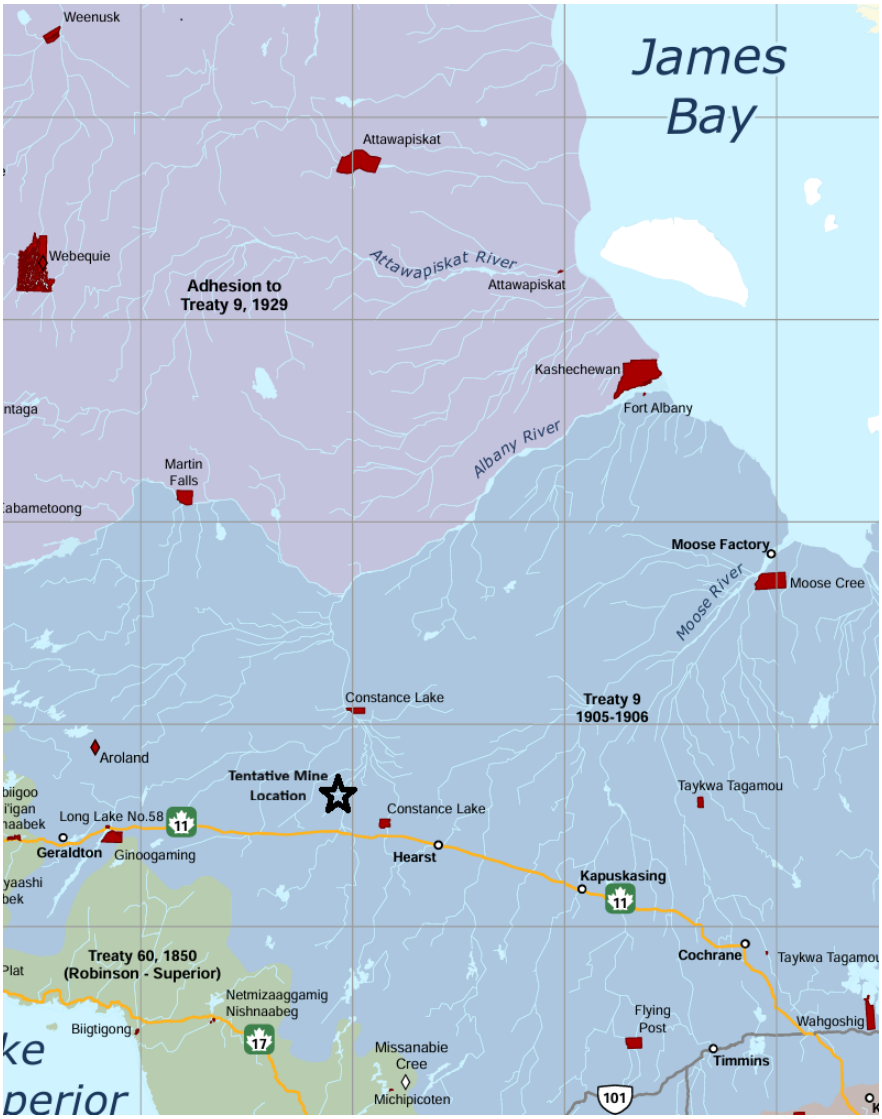


Figure B1. Map of First Nation Treaties and Indian Reserves near the proposed mine. [62]

Appendix C. Regulatory Stakeholders

Municipal Governments	**Provincial Government**	**Federal Government**
Municipal Council of Hearst	• Ontario Land Tribunal (OLT), to appeal and conduct hearings regarding EAs • Ministry of Environment, Conservation and Parks • Ministry of Natural Resources • Ministry of Labour, Training and Skills Development • Ministry of Northern Economic Development and Growth • Ministry of Energy and Mines • Economic Development, Job Creation and Trade • Ministry of Transportation	• Impact Assessment Agency of Canada • Fisheries and Oceans Canada • Health Canada • Indigenous Services Canada • Natural Resources Canada • Environment and Climate Change Canada

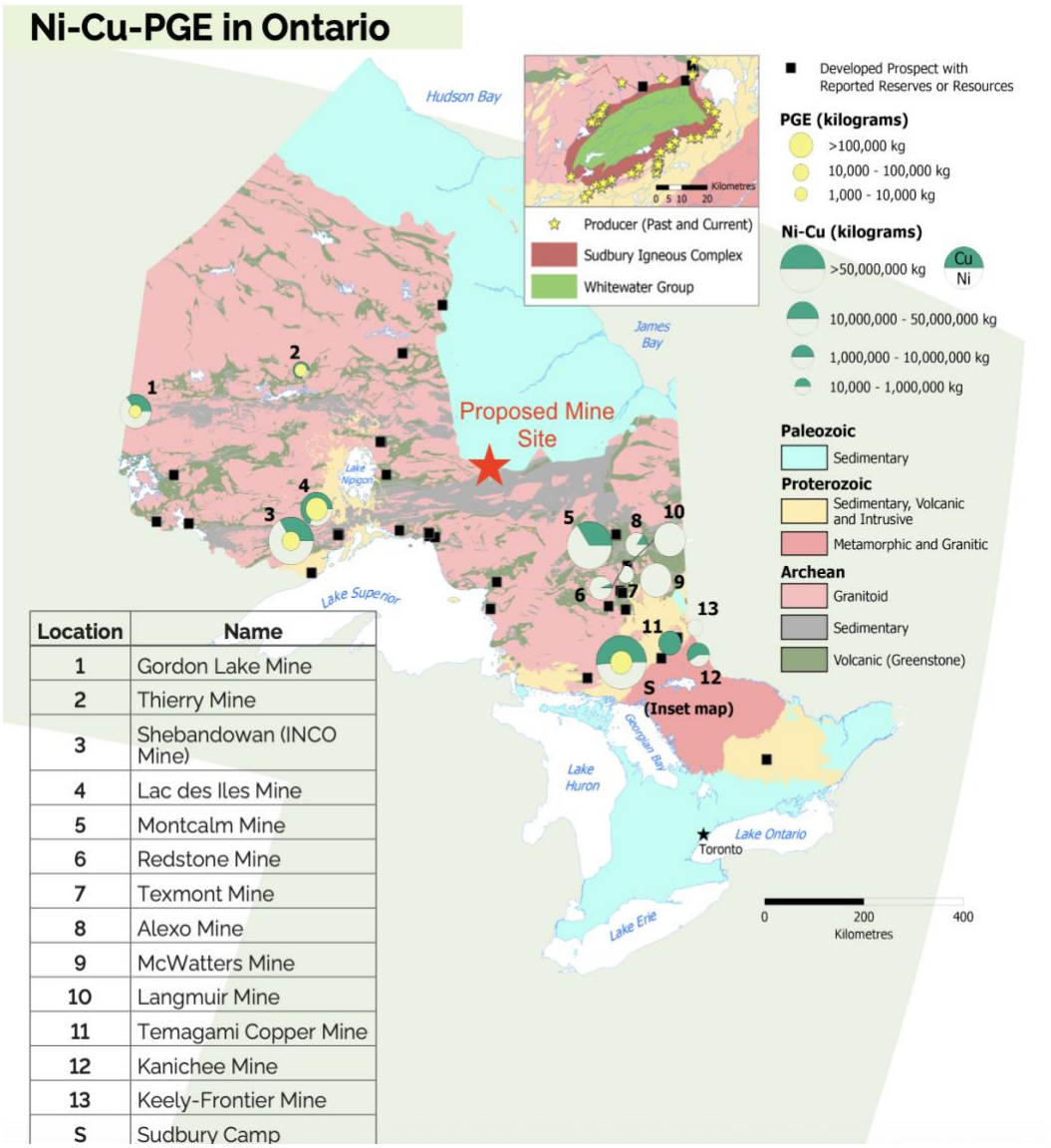
Table C1. Municipal, Provincial, and Federal Regulatory Stakeholders [61][63][63]

Appendix D. Minerals around Mining Site



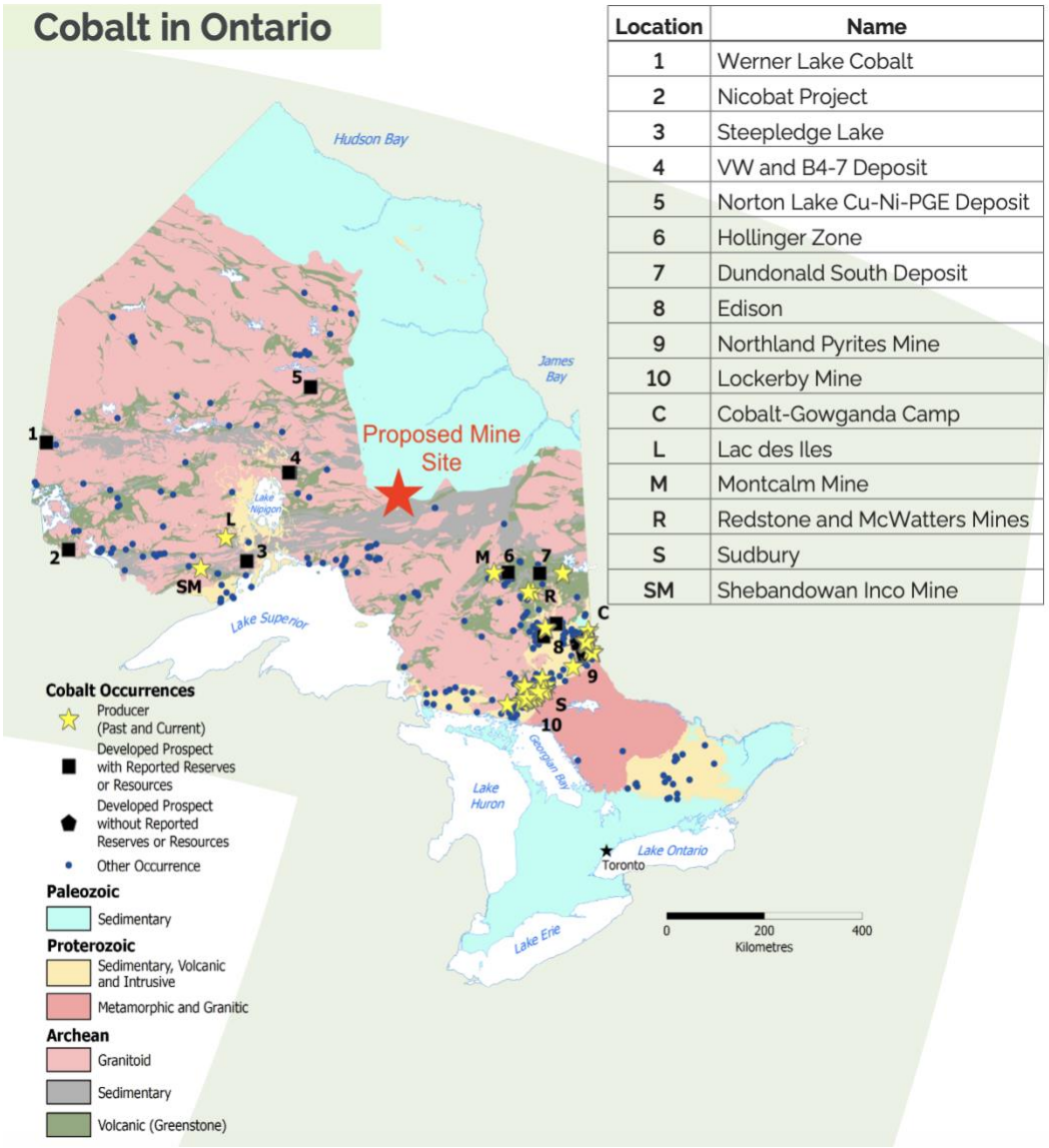
Location	Name
1	Big Mack and Separation Rapids
2	PAK and Spark Pegmatite
3	Mavis Lake
4	Wisa Lake North Zone
5	International Lithium
6	Raleigh Lake
7	Root Bay and McCombe
8	Rubellite and Pollucite Dikes
9	North and South Aubry
10	Georgia Lake area pegmatites

Figure D1. Map of known lithium deposits in Ontario [64]



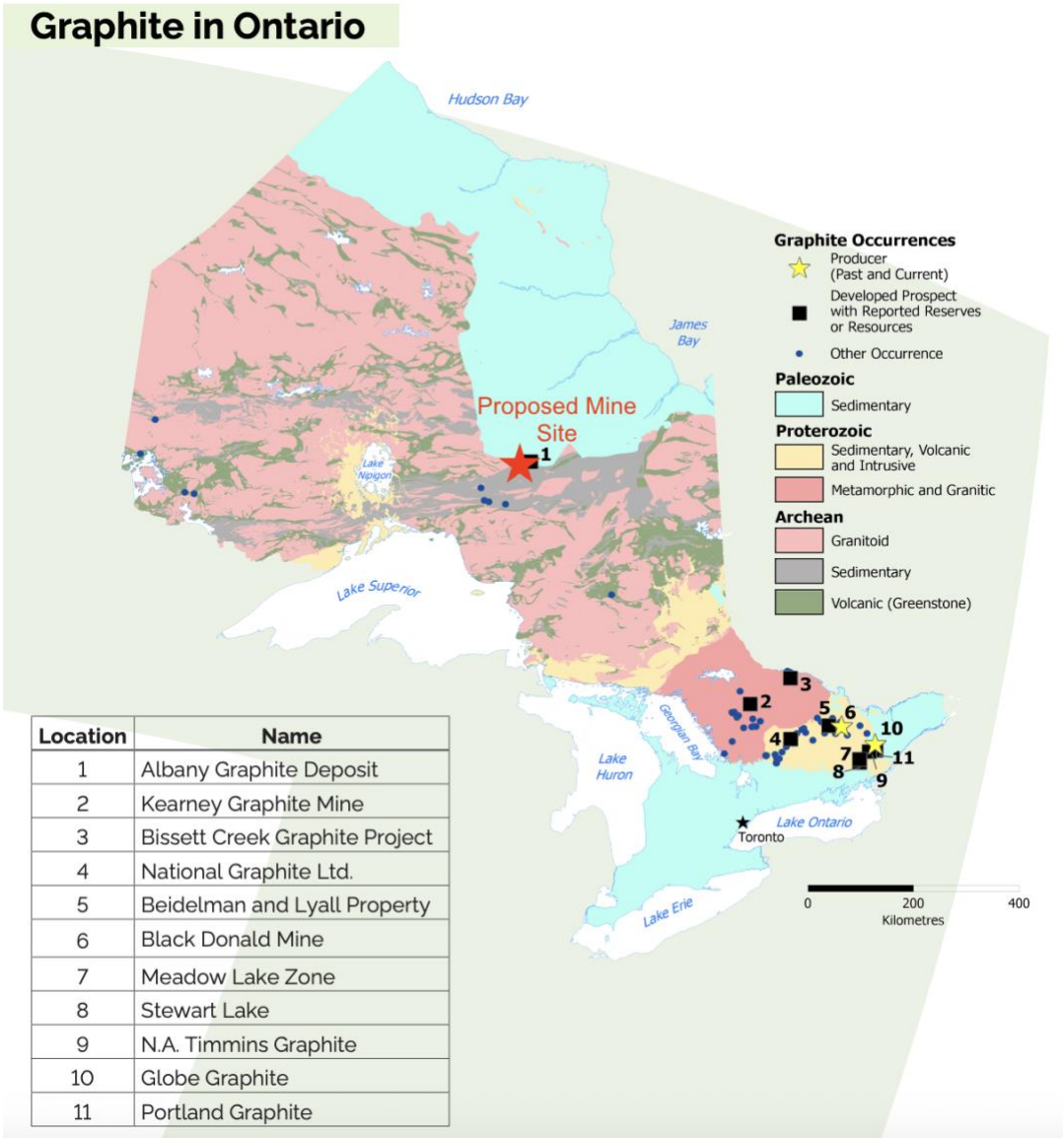
Location	Name
1	Gordon Lake Mine
2	Thierry Mine
3	Shebandowan (INCO Mine)
4	Lac des Iles Mine
5	Montcalm Mine
6	Redstone Mine
7	Texmont Mine
8	Alexo Mine
9	McWatters Mine
10	Langmuir Mine
11	Temagami Copper Mine
12	Kanichee Mine
13	Keely-Frontier Mine
S	Sudbury Camp

Figure D2. Map of known nickel deposits in Ontario [65]



Location	Name
1	Werner Lake Cobalt
2	Nicobat Project
3	Steepledge Lake
4	VW and B4-7 Deposit
5	Norton Lake Cu-Ni-PGE Deposit
6	Hollinger Zone
7	Dundonald South Deposit
8	Edison
9	Northland Pyrites Mine
10	Lockerby Mine
C	Cobalt-Gowganda Camp
L	Lac des Iles
M	Montcalm Mine
R	Redstone and McWatters Mines
S	Sudbury
SM	Shebandowan Inco Mine

Figure D3. Map of known cobalt deposits in Ontario [66]



Location	Name
1	Albany Graphite Deposit
2	Kearney Graphite Mine
3	Bissett Creek Graphite Project
4	National Graphite Ltd.
5	Beidelman and Lyall Property
6	Black Donald Mine
7	Meadow Lake Zone
8	Stewart Lake
9	N.A. Timmins Graphite
10	Globe Graphite
11	Portland Graphite

Figure D4. Map of known graphite deposits in Ontario [67]

Appendix E. Leopold Impact Matrices

Table E1. Leopold impact matrix for environmental valued components

Activity/VC	Caribou	Soil quality	Water quality	Vegetation	GHG emissions
Drilling	7/7	10/8	7/5	4/5	3/4
Excavation/ blasting	8/7	8/8	7/7	9/5	3/4
Land clearing	9/2	8/5	8/7	9/9	1/8
Construction	7/8	5/5	8/7	5/5	3/4
Mineral handling	3/3	2/2	10/8	5/5	3/4
Mineral transportation	4/3	2/2	7/5	3/2	3/4
Total	38/30	35/30	47/39	35/31	22/28

Table E2. Leopold impact matrix for social valued components

Activity/VC	Navigation	Essential services	Transportation	Residential communities/housing
Drilling	1/1	1/3	2/3	1/1
Excavation/blasting	1/1	2/4	4/6	1/1
Land clearing	5/7	3/5	7/7	5/7
Construction	5/7	5/6	6/8	3/6
Mineral handling	1/1	1/3	3/5	1/1
Mineral transportation	1/1	5/7	7/9	1/2
Total	14/18	17/28	29/39	12/18

Table E3. Leopold impact matrix for economic valued components

Activity/VC	Employment	Public utilities
Drilling	-4/-5	3/5
Excavation/blasting	-4/-5	3/5
Land clearing	-3/-2	4/4

Construction	-8/-3	5/7
Mineral handling	-4/-5	5/6
Mineral transportation	-4/-4	3/4
Total	-27/22	23/31

Table E4. Leopold impact matrix for health valued components

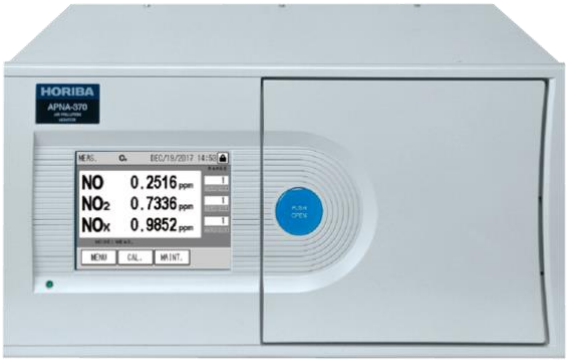
Activity/VC	Air quality	Noise and vibration
Drilling	7/7	5/6
Excavation/blasting	10/7	7/6
Land clearing	6/9	4/7
Construction	7/7	5/6
Mineral handling	7/6	3/5
Mineral transportation	7/7	5/6
Total	58/56	37/47

Table E5. Scale used for determining Leopold impact matrix values

Magnitude scale (-10 to 10)	Impact on stakeholders	Importance scale (-10 to 10)	Duration of benefit/ mitigation extent
-8 to -10	High positive	-8 to -10	Long-term benefit
-4 to -7	Medium positive	-4 to -7	Duration of project life
-1 to -3	Low positive	-1 to -3	Short-term benefit
0	None	0	Can be avoided
1 to 3	Low negative	1 to 7	Can be minimized
4 to 7	Medium negative	8 to 9	Can be restored
8 to 10	High negative	10	Can only be compensated

Appendix F. Example of Contaminant Monitors

Table F1. Commercially available ambient monitors from [69]

Contaminant	
CO	 Figure F1. APMA-370 Ambient Carbon Monoxide Monitor
NOx	 Figure F2. APNA-370 Ambient NOx monitor

SO2	 Figure F3. APSA-370 Ambient Sulfur Dioxide Monitor
PM2.5 and PM10	 Figure F4. APDA-372 Ambient Dust Monitor

Appendix G. Ontario's Ambient Air Quality Criteria (AAQC) for Contaminant Concentrations of Interest

Table G1. AAQC for contaminants of interest taken from [68]

Contaminant	AAQC ($\mu g/m^3$)	Averaging time	Notes
$PM_{2.5}$	27	24 hours	
PM_{10}	50	24 hours	
CO	15,700 (13 ppm)	8 hours	
NO_2	200 (0.10 ppm)	24 hours	Applicable to NO_X
SO_2	106.4 (40 ppb)	1 hour	

Appendix H.

Table H1. Maximum Concentrations of Deleterious Substances in Mining Effluent Water [44]

	Column 1	Column 2	Column 3	Column 4
Item	Deleterious Substance	Maximum Authorized Monthly Mean Concentration	Maximum Authorized Concentration in a Composite Sample	Maximum Authorized Concentration in a Grab Sample
1	Arsenic	0.10 mg/L	0.15 mg/L	0.20 mg/L
2	Copper	0.10 mg/L	0.15 mg/L	0.20 mg/L
3	Cyanide	0.50 mg/L	0.75 mg/L	1.00 mg/L
4	Lead	0.08 mg/L	0.12 mg/L	0.16 mg/L
5	Nickel	0.25 mg/L	0.38 mg/L	0.50 mg/L
6	Zinc	0.40 mg/L	0.60 mg/L	0.80 mg/L
7	Suspended Solids	15.00 mg/L	22.50 mg/L	30.00 mg/L
8	Radium 226	0.37 Bq/L	0.74 Bq/L	1.11 Bq/L
9	Un-ionized ammonia	0.50 mg/L expressed as nitrogen (N)	Not applicable	1.00 mg/L expressed as nitrogen (N)

Appendix I. Water Quality Monitoring Parameters

Table I1. Surface Water Quality Monitoring Parameters, Method, and Frequency of Sampling [44].

Parameters	**Method**	**Frequency**
Flow rate pH	Flow meter pH meter	Continuous
Total suspended solids	automatic sampler and laboratory analysis (24-hour composite)	3 times per week
Other deleterious substances (Arsenic, Copper, Cyanide, Iron, Nickel, Lead, Zinc, Radium 226, un-ionized ammonia)	Automatic sampler and laboratory analysis (24-hour composite)	Every week
Acute lethality (Daphnia magna, Rainbow trout)	Sample from the bodies of water and laboratory analysis	Every month (reduced to quarterly is 12 consecutive months of tests show no acute lethality)